>> Intelligent Systems – WC 7 <<

Machine Learning 2 and extra preparation for 2nd partial exam

Try to make exercise before you come to WC. By this means you will be aware of your own knowledge about the topics. Good luck!

Also there is a good chance that there will not be enough time to go through all the exercises during the working group meetings. Make sure you know what questions to ask.

Machine Learning 2

Exercise 1: Design Matrixes

Consider the following training set of positive examples {(1,1),(1,-1),(-2,2),(-2,-2)} and negative examples {(3,3),(3,-3),(-4,4),(-4,-4)}. Draw the feature space and try to fit a linear separator between the classes.

What can you do to rectify the problem? Provide a simple design matrix with two combined features instead of the original ones for which there is a simple separator, and give an example for such a linear separator.

Exercies 2: Linear Regression

Here we are going to practise fitting a linear function to a number of training sets. Suppose we have three pairs of values (1,1), (2,3) and (3,2) for two features. As discussed in the lecture, according to the ML recipe, the results depend on the chosen model class. Let us calculate the optimal function according to a number of different classes.

2a) Let us start with the most simple model class to fit our training examples, a constant line. This means that we are looking at a function f(x) = b, which is a horizontal line in the coordinate system. What is the result of learning the optimal function f(x) from the given data? Hint: we are looking for a value for b.

Solve this problem by using the following steps:

1. Calculate the value of the (squared) error function for each of the three points
2. Calculate the error function based on the points for the three points
3. Search for the minimal value for this function, by

a. Calculating the derivation of the function w.r.t. b

b. Setting this derivation to 0, and solve the equation for b.

The value you find for b gives you the function, i.e. the optimal horizontal line fitting your data.

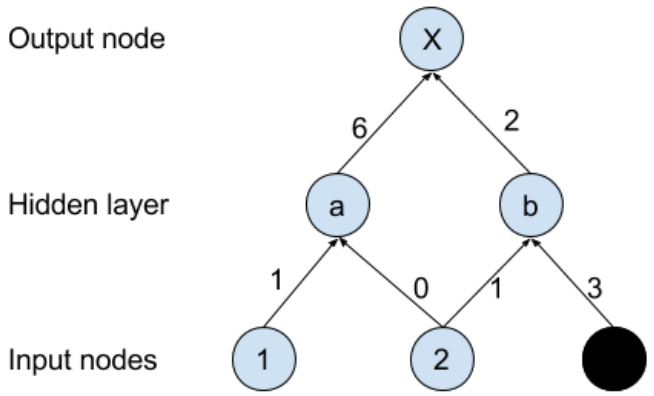
2b) A slightly more complex example is a line with a given, constant slope. Here we choose e.g. s=1. This means that we are looking at a function f(x) = 1*x + b, which is a line in the coordinate system with a slope of 45 degree. What is the result of learning the optimal function f(x) from the given data? Hint: we are still only looking for a value for b.

Follow the same method as described for 2a.

Exercise 2c (Not tested in the exam) The most interesting case for this simple example is, of course, when we also want to find the optimal slope s, i.e. a full linear function f(x)=s*x+b. The procedure is the same, only that the error function now also depends on s. This means you will have to calculate two partial derivatives, one for s and one for b. Setting both of them to be 0 will give you the optimal values for s and b. Try it out, but working with partial derivatives will not be tested in the exam.

<u>Exercise 3: Neural Networks</u>

Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?



Recap and Exam Preparation

<u>Exercise 4: Fuzzy Logic</u>

Given the following (standard) semantics for Fuzzy Logic for the following operators:

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A),f(B))
- f(A & B) = min(f(A),f(B))
- f(A -> B) = min(1;1 - f(A) + f(B))

What would be the fuzzy interpretation of the formula ((P -> ¬ Q) -> (Q v ¬P) given fuzzy values f(P)=0.9 and f(Q)=0.2 for P and Q.

<u>Exercise 5: Fuzzy Relations</u>

Consider the following relations, where WGS-5 and WGS-6 stands for Working Group Session 5 and Working Group Session 6, respectively.

Understanding:

	WGS5	WGS6
Joana	0.3	0.7
Andreea	0.6	0.4
Francesca	0.8	0.2

Contains;

	Fuzzy Logic	Probability
WGS-5	0.7	0.3
WGS-6	0.4	0.6

Fill in the following table for the composition of the two relations:

	Fuzzy Logic	Probability
Joana		
Andreea		
Francesca		

<u>Exercise 6: Calculating with Joint Probability distributions</u>

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

7a) P(cavity, toothache | catch) =
7b) P(-toothache | cavity) =
7c) P(-cavity, toothache, catch) =

<u>Exercise 7: Bayesian classifier</u>

There is a clear correlation between the weather, whether the NS operates smoothly and my level of frustration. The following table summarises some observations on five different days:

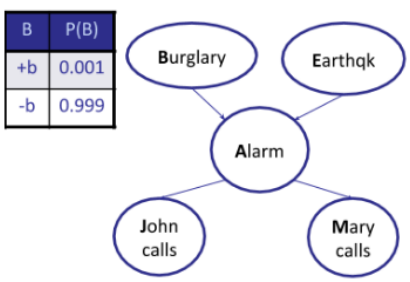
	Cold	Trains	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No

D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

Apply Naive Bayesian learning with a naïve smoothing of 0.01 on this data to predict whether I will be frustrated on a day where: it is not cold and no train is cancelled although it is windy and it snows.

Exercise 8: Bayesian Networks

Given the following Bayesian network:



B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

What is the probability that both John and Mary call, given that the alarm rings?

Advanced: What is the probability that Mary calls?

Exercise 9: Classification Errors

Given the following confusion matrix (system A):

	actual +	actual -
predicted +	100	20
predicted -	10	30

What is the Precision and Recall for this classification?

Exercise 10: Text feature preparation

What will the following sentences become after removing punctuation, stemming and stopwords?

a) Lisa is a student working on Artificial Intelligence.

b) Shuai was feeling lonely.

c) The intelligent systems course had a TA and many students.

Exercise 11: k-NN classification

Sentence a and c from the previous Exercise 10 are from the Canvas. Sentence b is from a diary. Let us use these examples as our training sets for a classifier that predicts whether a sentence is from Canvas or a diary. Considering the following sentence:

There are 200 students enrolled in the Intelligent Systems course.

Using K-NN classification with hamming distance, what are the two nearest neighbours for this sentence? And thus, what shall be the label if you need to predict the source of this sentence?